Math 162
Midterm Exam 1 solutions

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1. (a) Let $P = \{x_0, x_1, \cdots, x_n\}$ be a partition of $[0, 2]$. Since the irrational numbers are dense in $\mathbb{R}$, for any $1 \leq i \leq n$ there exists $x \in [x_{i-1}, x_i]$ such that $x \notin \mathbb{Q}$. Then we have $f(x) = 0$ and hence $m_i \leq 0$. On the other hand $f(x) \geq 0$ for all $x \in [0, 2]$, so $m_i \geq 0$. Thus $m_i = 0$ for all $1 \leq i \leq n$, which implies that $L(f, P) = 0$.

   (b) Letting P be a partition as in (a), we claim that for any $1 \leq i \leq n$, we have $M_i = x_i^2$. Clearly $M_i \leq x_i^2$, since the function $g(x) := x^2$ is increasing on $[0, 2]$ and hence $f(x) = g(x) \leq x_i^2$ for $x \in [x_{i-1}, x_i] \cap \mathbb{Q}$, while $f(x) = 0 \leq x_i^2$ for all $x \notin \mathbb{Q}$. On the other hand, if $x_i \in \mathbb{Q}$ then $f(x_i) = x_i^2$ and hence $M_i \geq x_i^2$ in that case. If $x_i \notin \mathbb{Q}$, we proceed as follows: fix $\epsilon > 0$, so by continuity of g there exists $\delta > 0$ such that $|x - x_i| < \delta$ implies that $|x^2 - x_i^2| < \epsilon$. By the density of $\mathbb{Q} \subset \mathbb{R}$, there exists $x \in \mathbb{Q}$ such that $x_i - \delta < x < x_i$, which implies that

   $$x_i^2 - f(x) = x_i^2 - x^2 < \epsilon.$$

   That is, we have shown that for any $\epsilon > 0$, there exists $x \in [x_{i-1}, x_i]$ such that $f(x) > x_i^2 - \epsilon$. It follows that $M_i \geq x_i^2$ in this case as well, which implies that that $M_i = x_i^2$ as claimed.

   As mentioned above, the function g is increasing on $[0, 2]$ and hence

   $$U(g, P) = \sum_{i=1}^{n} x_i^2 \Delta x_i = U(f, P).$$

   But we computed the integral of g in class, so we conclude that

   $$\inf\{U(f, P)\} = \inf\{U(g, P)\} = \int_0^2 x^2 \, \mathrm{d}x = \frac{8}{3}.$$

   (c) Since $L(f, P) = 0$ for any partition P, we have $\sup\{L(f, P)\} = 0$. Thus

   $$\sup\{L(f, P)\} = 0 \neq \frac{8}{3} = \inf\{U(f, P)\},$$

   which means that f is not integrable.

2. Note that $-|f| \leq f \leq |f|$, and hence

   $$-\int_a^b |f(x)| \, \mathrm{d}x = \int_a^b -|f(x)| \, \mathrm{d}x \leq \int_a^b f(x) \, \mathrm{d}x \leq \int_a^b |f(x)| \, \mathrm{d}x$$

   by #3(a) from Problem Set 2. This is equivalent to the desired inequality

   $$\left| \int_a^b f(x) \, \mathrm{d}x \right| \leq \int_a^b |f(x)| \, \mathrm{d}x \, .$$

3. (a) False. For example, if f is as in #1 and P is a partition such that $x_i \notin \mathbb{Q}$ for some $1 \leq i \leq n$, then $M_i = x_i^2$ but $f(x_i^*) \neq x_i^2$ for any $x_i^* \in [x_{i-1}, x_i]$. Thus in this case the i^{th} coefficient of $U(f, P)$ does not equal of the i^{th} coefficient of $S(f, P, \tau)$ for any tagging τ of P.

(b) False. For example, let $f : [0,1] \to \mathbb{R}$ be defined by

$$f(x) := \begin{cases} \frac{1}{x} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0. \end{cases}$$

Then f is unbounded and hence not integrable, but f is continuous everywhere except for the single point $x = 0$.

(c) True. Define $f : [0,1] \to \mathbb{R}$ by

$$f(x) := \begin{cases} 0 & \text{for } x \neq 0 \\ 1 & \text{for } x = 0. \end{cases}$$

Then $\int_0^1 f(x)\,\mathrm{d}x = 0$ by #3 on Problem Set 1.

(d) False. A counterexample is given by $f(x) = \sqrt{x}$ on $[0, \infty)$, which is unbounded and also uniformly continuous by #3(a) on Problem Set 3.